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Comparison of Bayesian seemingly unrelated regressions in high dimension.

Auteur : Gabriel DEGOIX

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Maître de stage : Zhi Zhao

University of Oslo, Department of Biostatistics, Domus Medica, Sognsvannsveien
9, 0372 Oslo, Norway

Enseignant référent ENSTA : Zacharie Ales

UMA ENSTA Paris, 828 boulevard de Maréchaux, 91120 Palaiseau, France

Note de confidentialité

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Abstract

Finding relationships between components in high-dimensional datasets is of great interest in many fields. To identify such relationships, [Zhao et al. \(2021\)](#) developed the BayesSUR R package, which provides users with several methods for predicting response variables based on predictors. These methods focus on Bayesian seemingly unrelated regression and offer different approaches for variable selection and error term estimation. Our objective is to compare these methods in a general context to gain perspective on the use of the package. This will be achieved by using simulated data from a broad simulation that we will conduct.

Keywords : high dimensional regression, Bayesian statistical learning, seemingly unrelated regression, variable selection

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1 Introduction

With the rapid advances in genome science and medicine, it is now possible to easily obtain a molecular characterization of a patient (Gray and Mills (2015)). We can consider for example a map of gene expressions. This type of data is particularly valuable for diseases like cancer, as we know, the disease can be linked to specific gene mutations. Today, with major breakthroughs in oncology, we aim to practice 'precision oncology'. Precision oncology refers to the study of cancer from a genomic point of view. With such precision, we can target specific gene expressions responsible for the disease without affecting others. This approach tends to be less toxic than more widespread treatments like chemotherapy, which acts on a much broader range of cells.

Since we know that certain gene expressions can be responsible for specific cancers, we aim to develop drugs that can effectively target and treat these conditions. To determine whether a drug has the desired effect on a cancer patient, we can conduct trials and analyze the data using linear regression. However, due to the vast amount of data involved in omics studies, it is not feasible to use such a simple model without first processing the data. This situation presents a high-dimensional problem that requires more sophisticated approaches.

In order to solve issue, the BayesSUR R package enables to process linear regression problems in high dimension using a Bayesian framework. It contains nine methods. These methods can also be use in other fields, e.g economics, ecology, finance, with similar model settings. We will use them for biology study here.

We will first understand what are the methods and how to tune them. Then, we will compare them on simulated data and draw some conclusions.

2 A short primer on Bayesian statistics

As we will use a Bayesian approach in this paper, here is a reminder of important principles from it.

2.1 Bayes' Rule

Bayesian statistics is an approach of statistics, often opposed to the frequentist approach, based on the Bayes theorem :

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

2.2 An example

One of the main difference between these two approach is how we are considering uncertainty for the variable of interest. We can take a simple example with the

coin flip game. We want to know the chance that the coin land on tails. Let note this event A.

Thus, how can we model this event ?

2.2.1 Frequentist approach

The frequentist approach will consider that the probability of A is π . Then, π is a number based on the specificity of the coin, the environment, etc. The probability we obtain is a fixed number, corresponding to the most likely one to be when regarding the data for example if we consider the maximum likelihood estimator. π represents the probability we will obtain when running the experiment an infinite number of time.

2.2.2 Bayesian approach

The Bayesian approach, however, considers that π is not a number but a distribution of probability. We are now having a more precise idea of what value the real π can take and with which probability. The idea behind Bayesian statistics is simple. Let the data talk. But with the possibility of incorporate some prior belief we have about the data. If you play with someone who you know uses cheated coins 90% of the time. Then if the first three flip are the same side, you could think with a greater probability than usual that he used a cheated coin. The idea is that with your belief on the data you can influence quicker the results. Nevertheless, if the coin is a regular coin and you run the experiment a lot of time, then you will still obtain a good probability that it is a regular one.

We could then define what form takes a prior. For example, when we want to modelize a Bernoulli probability, we want a distribution with support $[0, 1]$. Then, we can use a Beta distribution. For a Bernoulli probability, the Beta prior can be interpreted as follows : if $\pi \sim \text{Beta}(2, 1)$, that means that we construct our prior belief on the fact that we had 2 success and 1 fail. We can keep the same mean but gives more important on our belief by taking a $\text{Beta}(4, 2)$ distribution. Finally, we will not go further but the Beta prior is very interesting for computation because it is the conjugate prior for a Bernoulli probability. We always try to have conjugate priors when we can because it reduces a lot the computation as we then obtain an analytical form for the posterior.

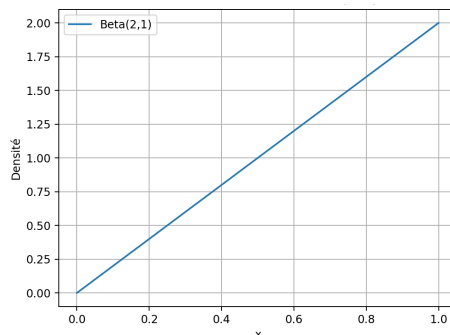
We can now take a simple example. Assume that we flipped the coin 3 times and had 2 tails. Then we can assume with what we know that π follows a distribution with support $[0, 1]$ and mean $2/3$. We can set a prior for π containing our knowledge. For example $\pi \sim \text{Beta}(2, 1)$. Let say we flip the coin 4 more times. The resulting event is D. Then, the new distribution of probability for π can be computed by :

$$P(A | D) = \frac{P(D | A)P(A)}{P(D)}.$$

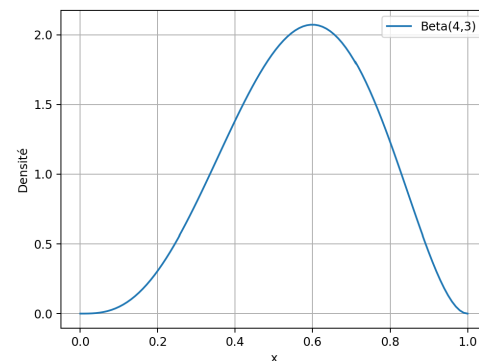
We can then identify known term and rewrite :

$$Posterior = \frac{likelihood \times prior}{normalisation}$$

If in D, we have 2 tails and 2 heads, regardless of the order it appeared, we have the posterior distribution for π being a $Beta(4, 3)$ (3 heads and 4 tails in total).



(a) Prior distribution



(b) Posterior distribution

We can see how with Bayesian we can "lead" faster our results to a space we want. Indeed, here with our prior knowledge we had an expected value of 0.67 for π . Then, even if in our data we obtained 2 tails and 2 heads after, the expected value has been pulled toward 0.5 but is still above. What happened is that we updated our beliefs on the coin. With the new data, the coin seems to be fair, then the mean of the distribution of π will get closer to 0.5. Nevertheless, because the size of the data we added is comparable to the one we had on our prior, we can see the new mean is between 0.55 and 0.6 which is coherent. We updated our belief on the coin "the data says it is a fair coin" but because we haven't done so much more tries to prove it we can still have some faith on our initial thought. In high dimensions problems as for genomics studies problem, as the space of probability can be huge, it can be very useful to guide our results with results from previous studies giving some knowledge on a situation (for example which genes are known to work together for a certain task).

Prior can be either informative or non informative. Non informative priors can help by giving a "method" to explore the data when informative priors are much more simple to understand as they just contains some information on the data as in our example.

Nevertheless, if the Bayesian framework allows so much advantages, it comes with it defects. Indeed, in some case, the computation of the posterior can be

difficult and take some times. Methods of MCMC are then used to approach posterior laws with a Markov Chain which needs to be stationary if we want to trust the results.

3 Bayesian seemingly unrelated regressions in high dimension

As we have a lot of genomics information from cancer patient, we want to know how to give them the best and most personalised treatment for their cancer. For different patient we can test several drugs which target different genes. We then want to know what is the relation between the drug response for the patients and the predictors which are the different genes in patients.

Because we are considering the responses of cancer patient to different drugs, we can legitimately believe that we are going to have some drugs which are targeting some genes in common and that we may have some correlations between the effects of drugs. We are going to make this assumption. Then, we are going to consider that we have no correlation between patients.

3.1 Formulation of the problem

In order to continue our example we take :

- n for the number of patient
- s for the number of drug tested
- p for the number of genes

Then this problem is simply a regression problem that we can write:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{U}, \quad (1)$$

$$\text{vec}(\mathbf{U}) \sim \mathcal{N}(0, \mathbf{C} \otimes \mathbf{I}_n). \quad (2)$$

Where $\mathbf{Y}$ is a $n \times s$ matrix giving the drug responses for every patient, its covariance matrix being $\mathbf{C}$. $\mathbf{X}$ is a $n \times p$ matrix for predictors. It contains the gene expressions of interest for each patient. $\mathbf{U}$ is the error term and $\text{vec}(\mathbf{U}) \sim \mathcal{N}(0, \mathbf{C} \otimes \mathbf{I}_n)$. That means that we take into account the correlation between drug responses into account (with $\mathbf{C}$) but that we consider no correlation between patient (with $\mathbf{I}_n$). Finally, $\mathbf{B}$ is $p \times s$ and is the regression matrix. It contains the relations we found between drugs and gene expressions.

We are searching for a relation between drug responses on cancer patient and their gene expression. Nevertheless, the number of genes and drug tested is bigger than the number of patient we have. We are on a high dimensional situation and we must reduce the space of predictors. In order to solve this issue, we will

consider the Bayesian variable selection model with a spike and slab prior used in the BayesSUR package. Then, to address the potential correlations between drugs, we will need to transform the equations we have in order to be able to do the computation. We will use for that Seemingly unrelated regression (SUR) factorization.

The variable selection is done with a latent indicator. These indicators are summarized in the Γ matrix. Then, this matrix, being the size of $\mathbf{B}$, will tell if β_{ij} is selected or not. A spike-and-slab prior as proposed in Lee et al. (2016) is used to recreate $\mathbf{B}$ from Γ . Then, conditional on $\gamma_{jk} = 0$, we have $\beta_{jk} = 0$ and conditional on $\gamma_{jk} = 1$ we have $\beta_{jk} \sim \mathcal{N}(\mathbf{0}, W_{\gamma}^{-1})$. Here, β_{γ} and W_{γ} are respectively the restriction of β and W conditional to $\gamma = 1$. Then, $W = w^{-1}\mathbf{I}_{sp}$ having a prior on w being : $w \sim \text{IGamma}(a_w, b_w)$. w should be tune based on the expected magnitude of effects for relevant covariates.

Then the spike and slab prior can be written :

$$\beta_{kj} \mid \gamma_{kj}, w \sim \gamma_{kj}\mathcal{N}(0, w) + (1 - \gamma_{kj})\delta_0(\beta_{kj}),$$

The BayesSUR package then allows us to use different priors for Γ and $\mathbf{C}$ independantly to each others. There is 3 priors for the covariance matrix and 3 for the Γ matrix which makes a total of 9 methods to find the $\mathbf{B}$ matrix. They are summarized in the following table.

	$\gamma_{jk} \sim \text{Bernoulli}$	$\gamma_{jk} \sim \text{Hotspot}$	$\gamma \sim \text{MRF}$
$\mathcal{C} \sim \text{indep}$	HRR-B	HRR-H	HRR-M
$\mathcal{C} \sim \text{IW}$	dSUR-B	dSUR-H	dSUR-M
$\mathcal{C} \sim \mathcal{HIW}_G$	SSUR-B	SSUR-H	SSUR-M

Table 1: Methods of BayesSUR

3.2 Priors for residual covariance

In the regression equation, we are looking at the error. In order to have a simple expression we vectorize the error matrix. Then, $\text{vec}(\mathbf{E})$ is following a normal distribution with for mean the null matrix and for variance the Kronecker product $\mathbf{E} \otimes \mathbf{I}_n$.

Here, $\mathbf{C}$ is the covariance matrix of the columns of $\mathbf{Y}$. Then, the Kronecker product with $\mathbf{I}_n$ makes a matrix like this :

$$\mathbf{C} \otimes \mathbf{I}_n = \begin{pmatrix} C & 0 & \cdots & 0 \\ 0 & C & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & C \end{pmatrix}$$

Then, it does not allow correlation between different people. We only admits correlations between different drugs for one person and then assume it is the same for the others.

The covariance matrix has 3 possible prior. It allows to specify the complexity of the model. Either we can consider the error is independent between responses variables or we can consider they are dependant. Then, there are 2 cases. The first, dSUR is using a transformation of the equations. The second one is considering a graph which makes the calculation faster and better.

3.2.1 Independent inverse gamma prior

The hierarchical prior is the most simple. It is considering that errors between response variables are independent from each others. That means that different drugs have no correlation between them. Every time, the error is sampled by a normal distribution centered on zero and with standard deviation σ_k for drug number k .

For this prior, the explicit form of the covariance matrix is :

$$\mathbf{C} = \begin{pmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_s^2 \end{pmatrix},$$

Then, we have priors for the σ_k^2 ,

$$\sigma_k^2 \sim \text{IGamma}(a_\sigma, b_\sigma).$$

That means we are going to sample every σ_k^2 from an Inverse Gamma distribution with parameters a_σ, b_σ . That's about the Bayesian framework we are in. We don't make any differences on the choices we are making for the standard deviation between drugs but we only assume that they comes from the same distribution that we are controlling through a_σ and b_σ . The fact that we allows these values not to be fixed with certainty allows a lot more flexibility than we have with the classic frequentist approach.

3.2.2 Inverse Wishart prior

Now with this prior for the covariance matrix, we assume correlation between the errors terms. Each regression equations for responses variable are as follows :

$$\mathbf{y}_j = \mathbf{X}\boldsymbol{\beta}_j + \mathbf{u}_j, \quad j = 1, \dots, s.$$

Then, the s equations seems independent but in reality, they are linked to each others by the $\mathbf{e}_j$ term. That is why we called this system of equations "Seemingly Unrelated Regression".

The dSUR prior is proposing an Inverse Wishart prior for the covariance matrix. A Wishart distribution is a generalization of the Gamma distribution for multivariate cases. A big advantage of this prior is that it is a conjugate prior for a covariance matrix from a multivariate normal distribution. That is our case. Thus,

$$\mathbf{C} \sim \mathcal{IW}(\nu, \tau \mathbb{I}_s).$$

Then, as the computation can be very expensive in Bayesian for this type of regression, we will use a transformation of the regressions equations proposed by Zellner and Ando (2010). The equations then become :

$$\begin{cases} \mathbf{y}_1 = \mathbf{X}\boldsymbol{\beta}_1 + \epsilon_1, \\ \mathbf{y}_2 = \mathbf{X}\boldsymbol{\beta}_2 + \rho_{21}\mathbf{u}_1 + \epsilon_2, \\ \vdots \\ \mathbf{y}_s = \mathbf{X}\boldsymbol{\beta}_s + \sum_{k=1}^{s-1} \rho_{sk}\mathbf{u}_k + \epsilon_s, \end{cases} \quad \epsilon_k \sim \mathcal{N}(\mathbf{0}, \sigma_k^2 \mathbf{I}_n).$$

Then, Bottolo et al. (2021), by exploiting this factorisation, had proposed the following priors for ρ_{kl} and σ_k^2 :

$$\begin{cases} \sigma_k^2 \sim \text{IGamma}\left(\frac{\nu-s+2k-1}{2}, \frac{\tau}{2}\right), \\ \rho_{kl} \mid \sigma_k^2 \sim \mathcal{N}\left(0, \frac{\sigma_k^2}{\tau}\right). \end{cases}$$

3.2.3 Hyper inverse Wishart (HIW) prior

The concept of the SSUR prior is the same as the dSUR adding that we will now use a graph in order to do our calculations. This graph corresponds is linked to the covariance matrix of $\mathbf{Y}$. In fact, the graph we will note $\mathbf{G}$ is the adjacency matrix of the covariance matrix of $\mathbf{Y}$, meaning it takes 1 if the term of the covariance matrix is non zero and zero otherwise. The $\mathbf{G}$ graph is then estimated by the SSUR method. We can select the sparsity of this graph with the η parameter. Then, we have a Bernoulli prior on the η probability to select an edge of this graph :

$$\eta \sim \text{Beta}(a_\eta, b_\eta).$$

That is something we are going to tune later.

Then, the method will consider for our calculations only the correlations corresponding to the vertices of this graph. That's what the "sparse" stands for. It is opposed to "dense", where we are considering every correlations for the model. The, we will consider an Hyper Inverse Wishart prior :

$$\mathbf{C} \sim \mathcal{HIW}_{\mathcal{G}}(\nu, \tau \mathbb{I}_s)$$

3.3 Priors for variable selection

Then, we 3 possible priors for $\mathbf{\Gamma}$ which are Bernoulli, hotspot and Markov random field.

3.3.1 Bernoulli prior

The Bernoulli prior is the most simple and understandable prior for variable selection. It can be written as follows :

$$\gamma_{jk} \mid \omega_j \sim \text{Ber}(\omega_j), \quad j = 1, \dots, p, \quad k = 1, \dots, s,$$

We assume here that, a priori, for every rows of $\mathbf{B}$ a variable has the probability ω_j to be selected. We can say that for every drugs we are testing, we assume at first that we have a probability (which is the same for a given drug) to have an effect on genes.

Then, ω_j is following a Beta prior : $\omega_j \sim \text{Beta}(a_\omega, b_\omega)$. This choice is coherent given that Beta distribution is the conjugate prior for a Bernoulli probability. In Bayesian statistics we are often using conjugate priors as it gives analytical posteriors that are easier to compute. A conjugate prior means that our posterior follows the same type of distribution than our prior.

3.3.2 Hotspot prior

The hotspot prior is similar to the Bernoulli prior. Nevertheless, the Bernoulli probability is no longer the same for a column but more specific to each variable. We can choose independently the given correlation between the predictors and the responses variables. That means we can choose two different probabilities for selecting a variable. One regarding which gene is considered and one regarding which drug is considered. It is offering more flexibility than the Bernoulli prior by giving a different probability to be selected for every variable. Then, we can easier identify some "hotspots" in $\mathbf{B}$. The idea is to have more flexibility.

This prior can be written :

$$\begin{aligned} \gamma_{jk} \mid \omega_{jk} &\sim \text{Ber}(\omega_{jk}), \quad j = 1, \dots, p, \quad k = 1, \dots, s, \\ \omega_{jk} &= o_k \times \pi_j, \\ o_k &\sim \text{Beta}(a_o, b_o), \\ \pi_j &\sim \text{Gamma}(a_\pi, b_\pi). \end{aligned}$$

As in [Zhao et al. \(2021\)](#), we can see the parameters composing ω_{jk} as : o_k accounting for the sparsity of each response model and π_j controlling the propensity of each predictor to be associated with multiple responses simultaneously.

3.3.3 Markov Random Field prior (MRF)

The MRF prior is a strongly informative prior that relate the relations between the predictors and the responses variables. That makes it different to the two first priors that are not informative and much more data driven. With an informative prior, we can have more understanding on how we are obtaining the results. The MRF prior works with a graph, containing the knowing relations between predictors and responses variables. In short, we can see it as a graph giving the information on whether some drugs as a known effect on some gene expressions or if we have correlation between drugs or between gene expressions. The MRF matrix can contain all the dependencies we know about the data what makes it very useful.

The density of this prior is:

$$f(\gamma \mid d, e, \mathbf{E}) \propto \exp\{d^T \gamma + e \gamma^T \mathbf{E} \gamma\}$$

This prior takes into account 2 things :

- The first part is proportional to $\exp\{d^T \gamma\}$. It represents the probability to select a variable as if they were all independent. This is the Bernoulli part of the prior.
- The second part takes into account the correlation between the predictors, the responses variables and both responses variables and predictors by the $\mathbf{E}$ graph.

Let explain quickly the construction of the $\mathbf{E}$ matrix ($ps \times ps$).

The graph is a Markov Random Field. It is a non oriented graph which gives the known dependencies between predictors with precision with which responses variables this dependencies is. To understand how the matrix takes into account these dependencies, we can take an example :

If we consider x_1, x_2, x_3 and y_1, y_2, y_3, y_4 . Assume that we know that x_1 and x_2 are associated with y_3 and y_4 . Then, we have the following shape for $\mathbf{E}$:

$$E = \begin{matrix} & \begin{matrix} \gamma_{11} & \gamma_{21} & \gamma_{31} & \gamma_{12} & \gamma_{22} & \gamma_{32} & \gamma_{13} & \gamma_{23} & \gamma_{33} & \gamma_{14} & \gamma_{24} & \gamma_{34} \end{matrix} \\ \begin{matrix} \gamma_{11} \\ \gamma_{21} \\ \gamma_{31} \\ \gamma_{12} \\ \gamma_{22} \\ \gamma_{32} \\ \gamma_{13} \\ \gamma_{23} \\ \gamma_{33} \\ \gamma_{14} \\ \gamma_{24} \\ \gamma_{34} \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix} \quad (8)$$

Now we know how to write the relations between predictors and responses variables in the matrix. Then, this task can be complicated. Then, when we will

design Σ , the covariance matrix of $\mathbf{X}$, we will be able to construct the matrix. With the form introduced by Zhao et al, 2021 [<https://doi.org/10.1093/jrsssc/qlad102>] we will be able to construct it easily. Indeed, it is necessary due to the huge size the $\mathbf{E}$ matrix can take.

The construction used in our R code is the same as the following example.

Let have x_1, x_2, x_3 correlated (we will have this relation in **Sigma**) with each other and with responses variables y_1, y_2 . Then we will construct this dependencies in the matrix with :

$$\underbrace{G_y}_{\text{for } y_1 \text{ and } y_2} \otimes \underbrace{G_x}_{\text{for } x_1, x_2 \text{ and } x_3} - \mathbb{I} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \otimes \mathbb{I}_3 - \mathbb{I}_8.$$

3.4 Posterior sampling with MCMC

To sample from the posterior distribution Zhao et al. (2021) uses the evolutionary stochastic search algorithm (ESS). Here, we run in parallel several tempered chains at the same temperature. Both exchanges moves and crossover can then happened between these chains. The mixing of these chains help to converge faster to the right space of the posterior distribution. These operations happen during the burning period when we try to get to the posterior. At the end of the burning period we then should have a stationary Markov Chain representing the posterior distribution. It is important for the Markov Chain to be stationary in order to have trust on the results. Indeed, that means we have good chances to have reached our goal to sample from the posterior. Otherwise, if the Markov chain is not stationary that means we definitely can't trust the results as the Markov chain is still "searching" for the space of posterior distribution.

The main chain samples from the un-tempered posterior distribution, which is used for all inference. Then, focus on the parameters we consider, we are using different samplers. The $\mathbf{B}$ coefficients are update at each iterations by a Gibbs sampler which sample from the posterior given above. Then, the γ_k $k = 1, \dots, s$, are updated from their conditional posterior using a Metropolis-Hastings sampler, jointly proposing an update from the corresponding β_k .

Finally, the $\mathbf{G}$ graph coefficients are updated using a junction tree sampler.

4 Hyperparameters tuning

4.1 Why?

In order to compare the different methods of the package, one of the main task is to find which hyperparameters gives the best results for each models. Indeed, we can compare methods only if we found the parameters that gives the best for each, otherwise we could choose a method where another one is better with a better

choice of hyperparameters.

For this, we are going to use different indicators but the most important ones will be WAIC (widely applicable information criterion) and LOO (leave-one-out) but without forgetting metrics like AUC and sensitivity which are very important in this kind of problem as we are searching to detect something.

4.2 How?

When we know what are each hyperparameters, we now need to understand well their functions. Then, we will make choice for their tuning. Indeed, we need to reduce the space of the possibilities for hyperparameters, otherwise, we will have infinite possibilities.

In our study, we will barely tune dSUR and HRR priors for the covariance matrix. Indeed, it can be difficult to interpret. Moreover, after some tries, we saw that they are not changing very much the results we obtain compared to the other priors. We will then focus on the tuning of $\mathbf{\Gamma}$ priors for and on the one for the η value, sparsity of the Hyper Inverse Wishart graph (for SSUR).

4.2.1 Bernoulli prior

The Bernoulli prior is the easiest to tune for variable selection. The idea is very simple. As it is representing the probability of selecting a variable considering every variables are independent from each other, we can think about it as the sparsity of the model. Then, the idea is to try different sparsity for the variable selection. The simple idea behind it and which will be used for the hotspot prior as well is the following. We know that :

$$\omega_j \sim \text{Beta}(a_\omega, b_\omega), \quad \text{for } j = 1, \dots, p.$$

Then,

$$\mathbb{E}[\omega_j] = \frac{a_\omega}{a_\omega + b_\omega}, \quad \text{for } j = 1, \dots, p.$$

In order to control the sparsity of the model, we can then fix a_ω to a certain value and take b_ω in consequence. That's what we did. Nevertheless, the possibilities are still infinite. Then, another way to further reduce the space of possible hyperparameters is to consider the following statement : we want to keep the benefits of flexibility offered by the Bayesian framework. In other words, we want the value of ω_j not to be too certain and close to the mean, it should be able to "navigate" close to the mean. Then, we simply want the variance on on this term to be small not to be too small.

A quick reminder, that for a given mean, the variance of ω_j following a Beta prior is :

$$\mathbb{V}[\omega_j] = \frac{\mathbb{E}[\omega_j](1 - \mathbb{E}[\omega_j])}{a_\omega + b_\omega + 1}, \quad \text{for } j = 1, \dots, p.$$

We can see that for a same expected value, the bigger is the value of the hyperparameters, the smaller is the variance. In fact we are giving all the more importance to our prior belief than to data when increasing the values of hyperparameters. We will then took small values for a_ω between 1 and 4.

4.2.2 Hotspot prior

The main idea behind the tuning of the hotspot prior is the same as for the Bernoulli prior. We set an expected value for the sparsity and we then play with the hyperparameters. Nonetheless, it is more difficult now as we have 4 hyperparameters. The space of possibilities is a lot bigger.

First, o_k and π_j being independent from each others, the expected value of ω_{ij} is :

$$E(\omega_{jk}) = E(o_k) \cdot E(\pi_j)$$

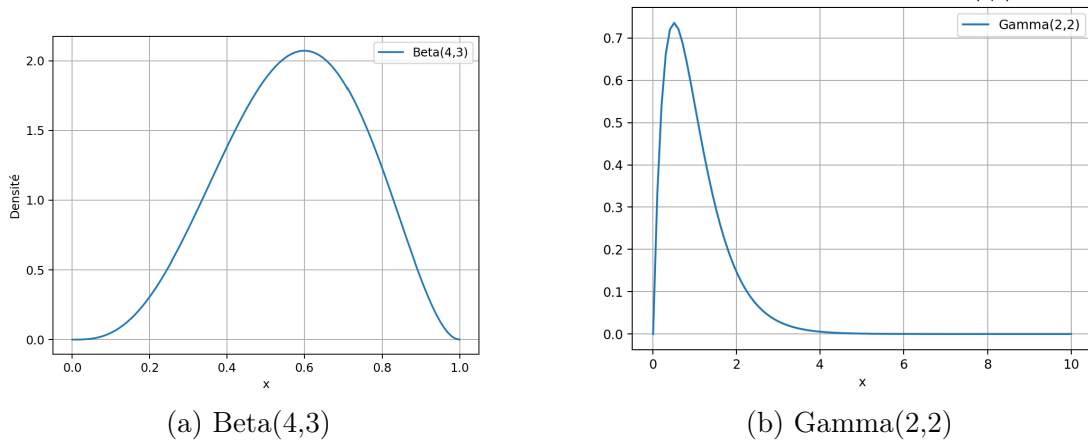
We can then tune separately the expected values of o_k and of π_j .

We know that o_k accounts for the sparsity of each responses models. Then, its value needs to be close to the mean of the column sparsity of Γ .

As we said, π_j accounts for the propensity of each predictor to be associated with multiple responses simultaneously. In their paper, [Richardson et al. \(2011\)](#) are explaining the effects of this parameter as giving the number of hotspot we can have. If π_j is big, we will have more propensity for hotspot in the j line and if π_j is small, then we will have fewer hotspot. Then, we choose π_j regarding the number of hotspot we want to have. Then, we will try different shapes for the Gamma distribution giving π_j with more or less weight close to the 0. We will then try different means from 0.8 to 2. An important point here is the value of the variance. Earlier, we said that we want the variance not to be too small in order to keep the flexibility offered by the Bayesian framework for this search. However, if it is true here for o_k , we must exercise greater caution with π_j . As it follows a Gamma distribution, its support has no upper bound. Then, even if we choose the mean, if the variance is too high, we can have ω_{ij} bigger than 1. That is why we need to have a compromise for the variance of π_j here.

The variance for π_j being :

$$\mathbb{V}[\pi_j] = \frac{\mathbb{E}[\pi_j]}{b_\pi}, \quad \text{for } j = 1, \dots, p.$$



4.2.3 MRF prior

The hyperparameters on the MRF prior that we have to tune are $\mathbf{d}$ and $\mathbf{e}$. $\mathbf{d}$ accounts for the sparsity of the model and $\mathbf{e}$ the strength we want to give to our knowledge on the dependencies summarized on $\mathbf{E}$. To tune these hyperparameters, we can follow some indication of the Lee et al. (2016) paper. First, we said $\mathbf{d}$ could represent the sparsity of the model. Lee et al explains that this sparsity can be written $d = \text{logit}(\Gamma_{\text{sparsity}})$. Then, a way to tune $\mathbf{d}$ can be the same as for the Bernoulli prior, choosing the sparsity of the model with d . We will note that the MRF prior with $\mathbf{e} = 0$ is equivalent to the Bernoulli prior with a certain value of ω_j . Then, to tune $\mathbf{e}$, Lee et al and Stingo et al advise to look at the "phase transition" we can detect when increasing the value of $\mathbf{e}$. It can be seen as the moment when model size is getting very big and a lot of variable are selected. Then, sensitivity is getting very high and specificity very low. Nevertheless, in our study we used both true graph and 25% true graph which seems to contains too much information to see that "phase transition" happens. In order to construct the 25% true graph we will delete uniformly the edges of each non-zero block. For example, if we have x_1, x_2, x_3, x_4 correlated with each others and related to y_1, y_2 , we will have for the 25% graph with only x_3, x_4 correlated and in relation with y_1, y_2 . We can see it on Figure 3 where the "small blocks" corresponding to the correlations between x 's are less important than before while we still have the "big block" corresponding to the correlations between the y 's.

To tune this prior, we will then try values $\mathbf{d}$ according to the wanted sparsity for the model. We need to consider a value of $\mathbf{d}$ a bit less of the sparsity desired giving that we will also give importance to the $\mathbf{E}$ graph. Then, for each values of $\mathbf{d}$, we will try $\mathbf{e}$ between 0 and 5.

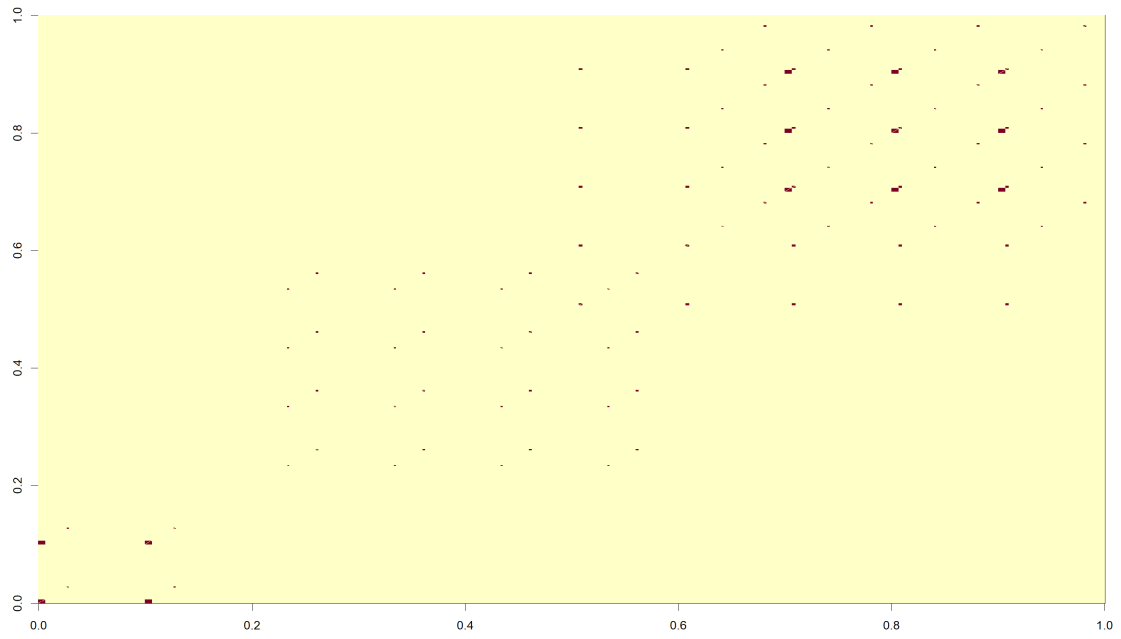


Figure 3: Complete E graph



Figure 4: 25% E graph

4.2.4 Graph of the HIW prior

For this prior, we will tune only the parameter η . It represents the sparsity of the $\mathbf{G}$ graph. Then, we can tune η the same way we tuned the Bernoulli prior for variable selection. We set an expected value for η and then consider different situation for the variable selection.

5 Simulation studies

5.1 Simulation scenario

For this study, we will be working with simulated data. Then, we will not have to estimate any quantities. The simulation is a very general simulation. We want to model here the response of cancer patient to different cancer drugs so we are in the case of biologic data. Nevertheless, our simulation scenario is much more general and can also model phenomenon on other fields as economics or marketing. Then, our results will be very general for the methods.

- Design a matrix of correlation Σ giving the correlation between gene expressions.
- Simulate the data $\mathbf{X} \sim \mathcal{N}(0, \Sigma)$, $\dim(\mathbf{X}) = n \times p$.
- Design a decomposable graph $\mathbf{G}$ containing the correlations between drugs, $\dim(\mathbf{G}) = s \times s$.
- Design a sparse matrix Γ giving the variable of importance, $\dim(\Gamma) = p \times s$.
- Simulate $\beta_{jk} \sim \mathcal{N}(0, 1)$, $j = 1, \dots, p$ and $k = 1, \dots, s$.
- Simulate $\tilde{U}_{ij} \sim \mathcal{N}(0, 1)$, $i = 1, \dots, n$ and $j = 1, \dots, p$.
- Simulate $P \sim \mathcal{W}_G(3, \mathbf{M})$ where diagonals of $\mathbf{M}$ are 1 and off-diagonals are 0.9, $\dim(P) = s \times s$.
- Use Cholesky decomposition $\text{chol}(P^{-1})$ to get $\mathbf{U} = \tilde{\mathbf{U}} \cdot \text{chol}(P^{-1})$.
- Generate $\mathbf{Y} = (\mathbf{X}\mathbf{B})_r + \mathbf{U}$.
- Tune the signal to noise ratio to be 10.

Here, we chose to take a signal-to-noise ratio of 10. We made this choice in order to not have too much noise but still enough to test the robustness of our methods. Then, we took $\mathbf{G}$ a decomposable graph composed of 3 blocs. That is why we have the parameter 3 in the Wishart distribution for P .

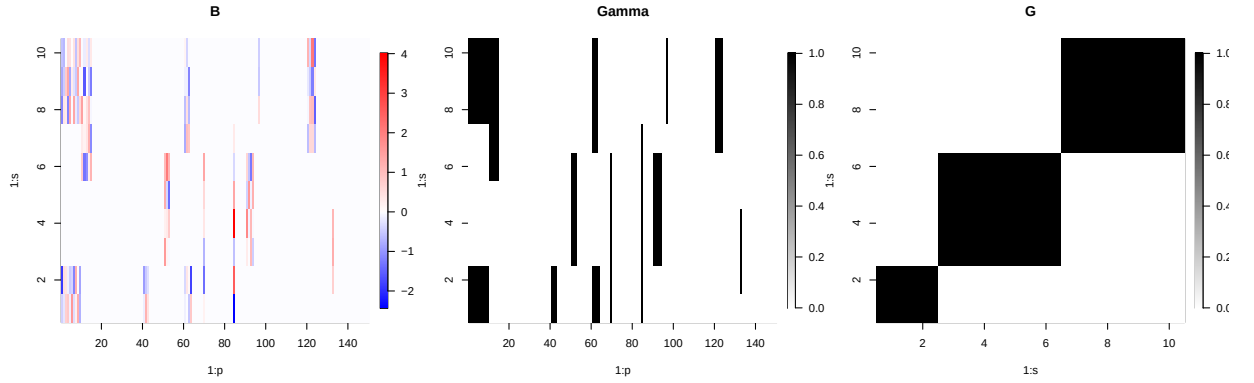


Figure 5: a0

5.2 Simulation results

To evaluate the different methods we are going to use several metrics. Some for the variable selection and some for the prediction.

For the variable selection we used classic metrics : accuracy, sensitivity, specificity and AUC.

We consider a true positive to be a selected variable in estimated Γ which is present in real Γ and a true negative to be a non selected variable in estimated Γ which is effectively not present in real Γ . Then, we can define the accuracy as the proportion of well classified variables, sensitivity as the true positive rate and specificity as the false negative rate.

- Accuracy = $\frac{TP+TN}{TP+FP+TN+FN}$
- Sensitivity = $\frac{TP}{TP+FN}$
- Specificity = $\frac{TN}{TN+FP}$.

The AUC is the name for Area Under the Curve. It refers to the ROC curve being the evolution of the sensitivity in relation to 1-specificity for a model. The bigger the area under the curve is, the better is the method. That basically means that we can have a good compromise between sensitivity and specificity.

For prediction, we used LOO, WAIC, RMSE and RMSPE. RMSE and RMSPE stands for Root Mean Square Error and Root Mean Square Predicted Error.

$$\text{RMSE} = \frac{1}{\sqrt{mn}} \|\mathbf{Y} - \mathbf{X}\hat{\mathbf{B}}\|_2$$

$$\text{RMSPE} = \frac{1}{\sqrt{mn}} \|\mathbf{Y}_{test} - \mathbf{X}_{test}\hat{\mathbf{B}}\|_2$$

RMSE is the training error. To calculate, RMSPE with our simulation scenario, we modified the random seed in our R code by adding a parameter a to the previous value. We did that 10 times before averaging the result. We then compute for our

estimation of RMSPE the standard deviation.

Then, we used WAIC and LOO which gives strong insights on the accuracy of the predictive model. We took it as presented in [Zhao et al. \(2024\)](#).

When we finished the tuning of hyperparameters, we could finally compare the results. They are summarized in the following table :

Table 2: Comparison of the methods

	AUC	accuracy	sensitivity	specificity	WAIC	LOO	RMSE	RMSPE	standard deviation
HRR-B	0.845	0.923	0.293	0.987	-993.176	-889.257	5.077	7.391	3.831
HRR-H	0.753	0.925	0.311	0.977	-1167.172	-1167.169	5.092	7.406	3.829
HRR-M	0.992	0.993	0.933	0.998	-853.253	-808.321	4.786	7.469	3.811
dSUR-B	0.962	0.947	0.518	0.990	-1339.912	-1328.279	4.518	6.989	3.940
dSUR-H	0.955	0.940	0.451	0.998	-1339.811	-1328.456	4.540	6.999	3.934
dSUR-M	0.998	0.995	0.957	0.999	-776.882	-766.150	5.752	6.926	3.958
SSUR-B	0.946	0.949	0.537	0.991	-1458.154	-1451.584	4.513	6.985	3.940
SSUR-H	0.946	0.943	0.476	0.999	-1457.744	-1451.340	4.553	7.014	3.939
SSUR-M									
Real graph	0.993	0.994	0.945	1.000	-1459.093	-1451.382	4.099	6.932	3.957
25% graph	0.960	0.969	0.713	0.998	-1457.897	-1450.931	4.107	6.940	3.956

Then, here are the matrix we obtained :

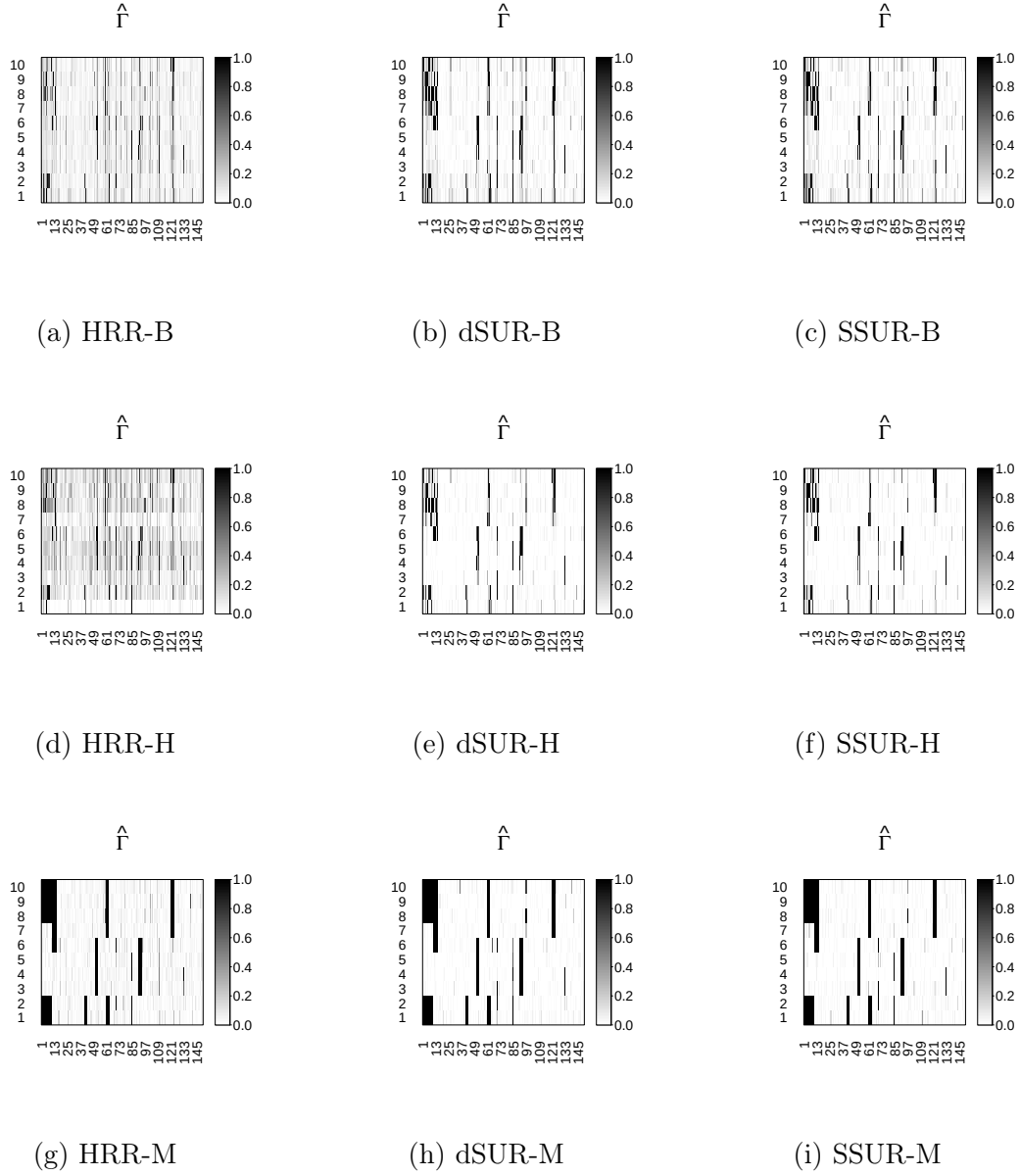
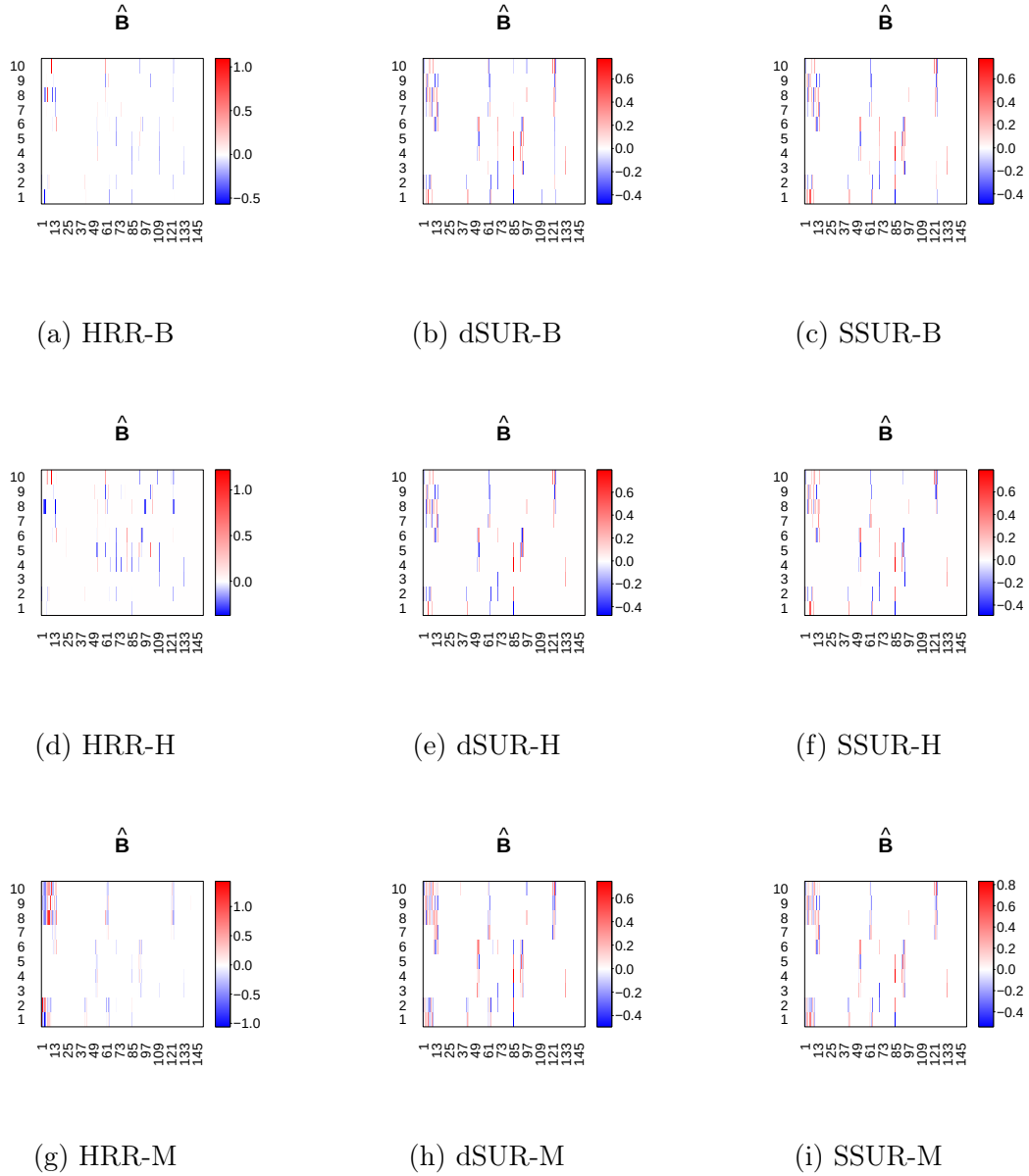


Figure 6: Estimated Γ matrix for the methods

Figure 7: Estimated $\mathbf{B}$ matrix for the methods

It is clear looking at the results that the MRF prior performed better than others regardless of the covariance prior chosen. This, even when we take only 25% of the graph. This shows the power of prediction of this strongly informative prior. Then, dSUR and SSUR models performed way better than HRR. Finally, dSUR and SSUR models have very similar results. This is probably due to the fact that the graph is not very sparse. Nevertheless, because of the $\mathbf{G}$ graph used for the SSUR prior, its computation time is much quicker which makes it a better method for our case.

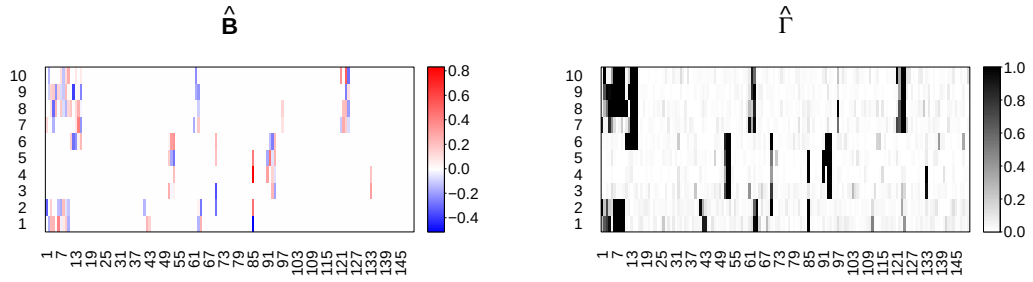
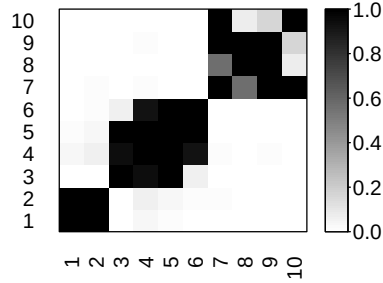


Figure 8: Estimators for SSUR-M with 25% graph

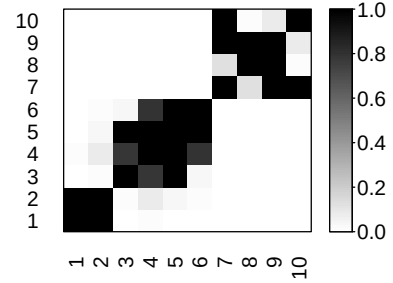
Then, we have summarized here the results for the estimation of $\mathbf{G}$ by the SSUR method. It is really well estimated, even when we have 25% of the graph as we can see with the metrics summarized in Table 3. The best set of hyperparameters in our case to tune the $\mathbf{G}$ prior were approximately the value of the true sparsity of $\mathbf{G}$ what makes sense. The fact that there is so much accuracy is also explaining why the SSUR methods works so well.

Table 3: Comparison of the $\mathbf{G}$ prior

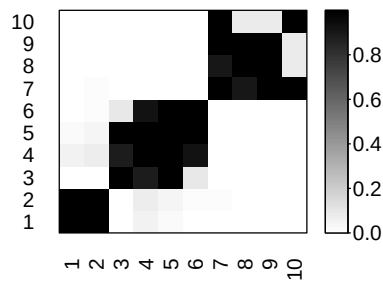
	accuracy	sensitivity	specificity
SSUR-B	0.94	0.83	1
SSUR-H	0.92	0.78	1
SSUR-M			
Real graph	0.94	0.83	1
25% graph	0.94	0.83	1

Estimated graph of responses

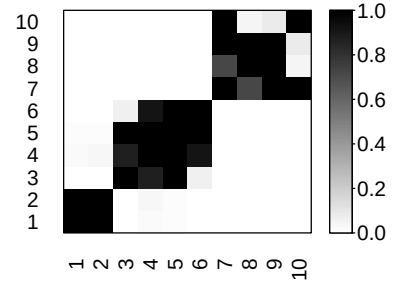
(a) Bernoulli prior

Estimated graph of responses

(b) Hotspot prior

Estimated graph of responses

(c) MRF prior

Estimated graph of responses

(d) 25% MRF prior

Figure 9: Estimated G graph

6 Description of contributions

Quick sentence to talk about how we compared methods

6.1 MRF prior understanding

One big difficulty I encounter during this internship has been to have a well understanding of the MRF prior. At first, I had difficulties to really understand how to construct the G matrix and when I first thought I understood it, I did a wrong matrix which led to non satisfying results. Indeed, we had really low sensitivity (true positive rate) for the method and a "phase transition" happening very quickly

and giving importance to some variables that were not of interest.

What happened then was that I was not correctly putting the prior knowledge on correlation between the predictors and response variables (contained in Σ and $\mathbf{G}$) in the $\mathbf{E}$ graph. It took me some time and reading before to correct it and have the right matrix at the end.

6.2 Hotspot prior tuning

The tuning for the hotspot prior has been particularly difficult. Because of the huge size of the space of hyperparameters we were testing, it took a long time to tune it. Moreover, we had to be careful on not to have ω_{jk} bigger than 1 what led us to study the shapes of the distribution to test for ϕ_k and π_j .

6.3 Use of AUC

As my work was to compare the different methods of the BayesSUR package, I began with the easiest ones to understand and the most simples (HRR-B and HRR-H). Nevertheless, I soon encounter problems regarding the sensitivity for these methods, which was very low and way below the results we wanted. In other words, they under-performed for detecting variables of interest in Γ comparing to other papers using the method. We then decided to move on to other methods as dSUR-M or SSUR-M that allows to take into account correlations in the errors and which have a strongly informative prior for the variable selection. These methods can be a bit more difficult to understand but offer better results. This made me more familiar with the BayesSUR package at the beginning.

Then, when I went back to the more "simple" methods, I still had a sensitivity very low and very high specificity (1 or 0.99). That meant that the method were unable to detect enough variable, that the sparsity of our estimated Γ was too low. This was happening even when taking a big expected value for ω_j in the case of methods with the Bernoulli prior. This last observation made us thought about how the AUC could be a metrics of interest here. With this metric, we can have a bigger picture of what is happening. If the AUC is satisfying but the sensitivity not, that means that the method is not sensitive for the threshold we use (0.5) but is sensitive for a lower one. The signal is weaker but the method can still be interesting.

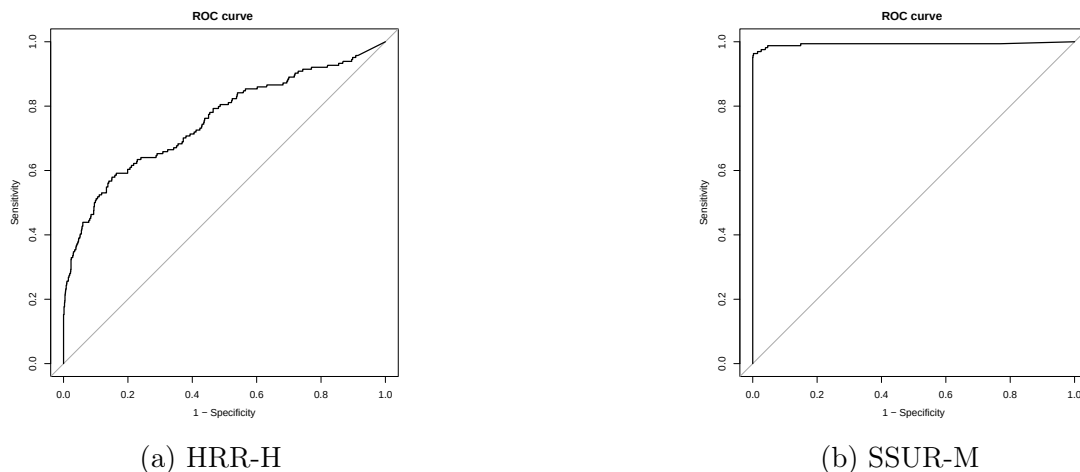


Figure 10: ROC curve obtained for 2 methods

6.4 Computation time and MCMC

A need of Bayesian statistics being to compute complicated posterior distributions, we often need to use MCMC algorithms. In our case, we were running between 3 and 6 Markov chains for 200 000 iterations and a burnin period of 100 000 iterations. Then, the computation time can be long. One simulation was about 10 to 20 minutes. With the huge number of hyperparameters to try for each method, I could not do it only by using my computer. I then had access to the server of the department in order to do the simulations. That allowed us to be more precise in the tuning and to explore a lot more options.

Then, an essential feature when using MCMC is to look at the MCMC diagnosis. Indeed, when running a MCMC algorithm in order to sample for the posterior, we can trust the results only if we obtain a stationary Markov chain at the end of the burnin period. If we not, the results we obtain worth nothing. That situation happened when tuning the HRR-B and HRR-H methods. We had non stationary Markov chain at first when we tried with 3 chains. We then tried to tune the number of chains to use and found that we had stationary results for 6 chains. Afterwards, we kept this new number of chains for the 2 methods.

Conclusion

Our study of the different comparison found that the MRF prior were very efficient for variable selection and offered the models to be very good. Nevertheless, if in our case this prior is so much powerful this is also because we have put a huge quantity of data knowledge in it. In the reality, we can have sometimes really few knowledge on the data and then the Markov random field is not much more a very informative prior. Indeed, when we have very few informations, then the second part of the prior containing the $\mathbf{G}$ become very sparse so negligible and only the first part of

the prior have an importance. The MRF prior then become nothing less than a Bernoulli prior with a certain value for the ω_{jk} probability (no variance). Then, other non-informative priors can be more effective. We can talk of the Hotspot prior which underperformed the Bernoulli prior in our case even if it allows for more flexibility. It can be explained because we don't have much hotspot in our simulated $\mathbf{\Gamma}$. The data is more scattered. Nevertheless, if we had big groups on our data like genes that works together, then the hotspot prior should have obtained better results than the Bernoulli prior when tuned well.

The main conclusion we can draw is that when we have a lot of informations on our data the Markov Random Field should outperform the two other methods. Then, when we have very few or no information on the data, we should try again to see which techniques work the most. Indeed, as we know in statistical learning, there is no such methods that outperformed all the others, only methods that works for some given problems.

On a personal plan, this internship taught me a lot about Bayesian statistics, a topic I had only a few courses on and that I understand way better now. It has also reinforced my knowledge on statistical learning and gave me another point of view in order to deal the high dimensional problem for regressions. What was the more profitable was to solve work with a real problematic making the understanding of the Bayesian methods used very clear.

Annexes

R code in the following Github : <https://github.com/gabi1771/BayesSUR-comparison>

References

- Bottolo, L., Banterle, M., Richardson, S., Ala-Korpela, M., Järvelin, M.-R., and Lewin, A. (2021). A computationally efficient Bayesian seemingly unrelated regressions model for high-dimensional quantitative trait loci discovery. Journal of the Royal Statistical Society: Series C (Applied Statistics), 70(4):886–908.
- Gray, J. W. and Mills, G. B. (2015). Large-scale drug screens support precision medicine. Cancer Discovery, 5(11):1130–1132.
- Lee, K. H., Tadesse, M. G., Baccarelli, A. A., Schwartz, J., and Coull, B. A. (2016). Multivariate bayesian variable selection exploiting dependence structure among outcomes: Application to air pollution effects on dna methylation. Biometrics, 73(1):232–241.
- Richardson, S., Bottolo, L., and Rosenthal, J. S. (2011). Bayesian models for sparse regression analysis of high dimensional data. Bayesian Statistics 9, pages 539–568.
- Zellner, A. and Ando, T. (2010). A direct Monte Carlo approach for Bayesian analysis of the seemingly unrelated regression model. Journal of Econometrics, 159(1):33–45.
- Zhao, Z., Banterle, M., Bottolo, L., Richardson, S., Lewin, A., and Zucknick, M. (2021). Bayessur: An r package for high-dimensional multivariate bayesian variable and covariance selection in linear regression. Journal of Statistical Software, 100(11):1–32.
- Zhao, Z., Banterle, M., Lewin, A., and Zucknick, M. (2024). Multivariate bayesian structured variable selection for pharmacogenomic studies. Journal of the Royal Statistical Society Series C: Applied Statistics, 73(2):420–443.